

TURUKHANSK, Russian Federation

WMO: 234720

Lat: 65.794N

Lon: 87.989E

Elev: 38

StdP: 100.87

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.7	-43.7	-48.9	0.0	-46.7	-46.1	0.0	-43.6	9.4	-15.3	8.7	-17.0	1.4	60	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	27.3	18.0	25.1	17.0	23.0	16.1	19.4	24.9	18.1	23.2	16.9	21.5	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.5	21.8	16.2	11.5	20.3	14.8	10.6	19.3	55.6	25.0	51.6	23.2	47.8	21.6	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	DB	-48.0	29.8	3.4	2.2	-50.5	31.4	-52.5	32.7	-54.4	33.9	-56.9	35.5	(4)
(5)			WB	-48.0	20.8	3.4	1.6	-50.4	21.9	-52.5	22.8	-54.4	23.7	-56.9	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.5	-26.5	-22.9	-13.6	-6.5	1.6	12.7	16.2	12.6	5.7	-4.3	-19.3	-23.2	(6)	
	DBStd	16.56	10.26	8.60	8.54	7.26	5.54	5.65	4.09	4.00	4.58	6.54	10.25	10.32	(7)	
	HDD10.0	6097	1131	921	732	494	267	37	2	17	143	444	878	1029	(8)	
	HDD18.3	8754	1389	1154	991	744	519	184	91	184	379	703	1128	1287	(9)	
	CDD10.0	430	0	0	0	0	7	117	195	97	14	0	0	0	(10)	
	CDD18.3	46	0	0	0	0	0	14	26	6	0	0	0	0	(11)	
	CDH23.3	392	0	0	0	0	2	130	227	33	0	0	0	0	(12)	
	CDH26.7	79	0	0	0	0	0	27	47	6	0	0	0	0	(13)	
(14) Wind	WSAvg	3.1	3.3	3.1	3.3	3.3	3.2	2.9	2.8	2.8	3.0	3.4	3.1	3.4	(14)	
(15) Precipitation	PrecAvg	603	32	30	33	34	43	55	62	82	65	76	49	42	(15)	
	PrecMax	795	70	65	76	66	83	145	122	130	190	124	90	88	(16)	
	PrecMin	424	9	6	5	7	14	11	3	21	25	42	15	9	(17)	
	PrecStd	107	15	14	15	17	17	34	31	33	39	25	19	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.6	-3.0	3.9	11.5	21.7	30.0	30.2	26.9	21.0	10.5	1.6	-1.6	(19)	
		MCWB	-4.1	-3.6	0.1	4.8	12.4	18.5	19.6	19.2	13.6	8.4	0.6	-2.2	(20)	
	2%	DB	-8.3	-5.9	1.1	8.0	15.7	27.0	27.8	23.5	16.9	6.7	-0.1	-3.5	(21)	
		MCWB	-9.1	-6.2	-0.7	3.2	8.6	16.8	18.5	16.9	11.7	5.3	-0.5	-4.0	(22)	
	5%	DB	-10.6	-8.7	-0.8	5.6	12.5	24.6	25.9	21.2	14.4	4.9	-2.1	-5.9	(23)	
		MCWB	-11.1	-9.3	-2.2	1.8	6.8	15.9	17.7	15.7	10.7	3.8	-2.6	-6.5	(24)	
	10%	DB	-13.1	-10.9	-2.3	3.4	9.4	22.1	23.9	19.0	12.3	3.2	-4.6	-9.0	(25)	
		MCWB	-13.5	-11.5	-3.4	0.6	4.9	14.7	17.0	15.0	9.9	2.0	-5.2	-9.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.0	-3.6	0.8	5.1	12.6	20.1	21.5	20.1	14.8	8.6	1.0	-2.0	(27)	
		MCDB	-3.6	-3.2	2.5	10.0	20.0	27.0	27.7	25.7	18.8	10.7	1.3	-1.8	(28)	
	2%	WB	-8.9	-6.2	-0.5	3.5	9.3	18.3	20.0	18.0	13.0	5.8	-0.5	-4.0	(29)	
		MCDB	-8.4	-5.9	1.0	7.1	14.7	24.7	25.2	21.5	16.0	6.5	-0.1	-3.5	(30)	
	5%	WB	-11.1	-9.3	-2.0	2.3	7.3	16.8	18.8	16.6	11.5	3.9	-2.6	-6.5	(31)	
		MCDB	-10.5	-8.8	-1.0	4.9	11.8	22.8	24.1	19.9	13.6	4.8	-2.1	-6.0	(32)	
	10%	WB	-13.5	-11.4	-3.4	1.0	5.3	15.5	17.7	15.3	9.9	2.0	-5.2	-9.5	(33)	
		MCDB	-13.1	-10.9	-2.3	3.4	8.8	20.9	22.5	18.4	12.3	2.9	-4.6	-9.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.1	9.8	9.9	7.6	8.6	8.7	7.0	5.6	4.7	6.9	6.9	(35)	
		MCDBR	7.1	9.3	7.5	8.4	10.9	12.4	12.2	10.3	8.8	4.5	5.7	7.5	(36)	
	5% WB	MCWBR	7.1	9.1	6.3	5.6	6.3	5.7	5.5	5.0	5.1	3.9	5.7	7.5	(37)	
		MCWBR	7.2	9.4	7.4	7.6	10.1	10.5	10.4	8.7	7.2	4.5	5.8	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.224	0.275	0.319	0.342	0.363	0.380	0.415	0.399	0.342	0.283	0.208	0.220	(40)	
		taud	1.858	1.985	1.994	2.036	2.157	2.387	2.272	2.288	2.402	2.174	1.891	1.866	(41)	
	Edn at Noon		193	554	713	805	831	830	779	750	721	557	203	23	(42)	
		Edh at Noon	21	63	103	127	125	103	113	100	71	51	19	6	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.10	0.73	2.22	4.18	5.33	5.48	4.94	3.13	1.77	0.67	0.19	0.02	(44)	
	RadStd		0.01	0.06	0.21	0.47	0.49	0.47	0.60	0.39	0.23	0.12	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.98	N/A	N/A	N/A	N/A	N/A	-372	-455	N/A	N/A
(47) Regional (2 neighbors)		+0.98	N/A	N/A	N/A	N/A	N/A	-372	-455	N/A	N/A

Nomenclature: See separate page